

1. The derivative of a function of one variable is defined as

$$\frac{df(x)}{dx} = f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

if the limit exists. Calculate the derivative of the following function using the above definition:

$$f(x) = \frac{x}{x-1}.$$

2. Calculate the gradient of the following function:

$$f(x_1, x_2, x_3) = e^{x_1} + x_1^2 x_3 - x_1 x_2 x_3.$$

3. Consider the function f , and calculate the directional derivative along the direction d at a point (x_1, x_2, x_3) , where

$$f(x_1, x_2, x_3) = e^{x_1} + x_1^2 x_3 - x_1 x_2 x_3, \quad (1)$$

$$d = \begin{pmatrix} d_1 \\ d_2 \\ d_3 \end{pmatrix}. \quad (2)$$

4. Consider the function

$$f(x, y) = x^2 + y^2.$$

The directional derivative of f in the direction of the vector $(1, 1)$ at the point $(0, 0)$ is $2\sqrt{2}$. Is it correct?

5. Calculate the steepest descent direction of the following function

$$f(x) = \frac{1}{2}x_1^2 + 2x_2^2,$$

at point $x = (1, 1)$.

6. Calculate the Jacobian matrix given the following function:

$$f(x_1, x_2) = (x_1^2 x_2, \cos(e^{x_1 + x_2^2})).$$

7. The surface of a lake is given by a region D and a plane XY . It's depth (in meters) in point (x, y) is given by the function:

$$f(x, y) = 400 - 3x^2 y^2.$$

If a swimmer is located in point $(1, -2)$, determine which direction he needs to swim in order to make the depth increase as quick as possible.